ABERDARE TECHNICAL COLLEGE

LEVEL 5,MODULE 3

CAT2,TERM 3

ELEMENT 2:TEST COMPUTER SOFTWARE FUNCTIONALITY

UNIT OF COMPETENCE: INSTALL COMPUTER SOFTWARE

UNIT CODE: 0619 451 06A

Total Marks: 36 **Time:** 1 Hour 30 Minutes

Name: _____ Reg. No: _____ Date: _____

Total Marks: 40 **Time:** 1 Hour 30 Minutes

QUESTION 1 (8 marks)

(a) Define the term software testing. **(2 marks)**

(b) Define each of the following terms: **(3 marks)**

(i) Error

(ii) Bug (defect)

(iii) Failure

(c) State any three HTTP methods used in REST and give the purpose of each. **(3 marks)**

QUESTION 2 (8 marks)

(a) Explain the difference between verification and validation, giving one example of each. **(4 marks)**

(b) Using the restaurant analogy (customer, waiter, kitchen), explain what an API is and how it works. **(4 marks)**

QUESTION 3 (8 marks)

(a) An age field on a registration form accepts values from 18 to 60 only. **(4 marks)**

(i) Use equivalence partitioning to identify the valid and invalid partitions. (2 marks)

(ii) Use boundary value analysis to list the six values you would test. (2 marks)

(b) A school website has the base URL https://school.com/api/. State the HTTP method and a suitable endpoint you would use in Postman to: **(4 marks)**

(i) Retrieve the list of students

(ii) Add a new student

(iii) Update a student's details

(iv) Delete a student record

QUESTION 4 (8 marks)

(a) Compare black box testing and white box testing under the following headings: knowledge of code, skills required, coverage, and what each type finds. **(4 marks)**

(b) A login form accepts passwords of 8 to 12 characters. Analyze how static testing and dynamic testing would each detect a missing check for the 12-character maximum, and explain why the two approaches complement each other. **(4 marks)**

QUESTION 5 (4 marks)

A bank is building a new system that must handle financial transactions with strong built-in security, while a sister company is building a public mobile app backend for simple data retrieval.

Evaluate which of REST or SOAP is better suited to each project. Justify your choices with at least two reasons per project. **(4 marks)**

(